Derivation of Expected Shortage per Replenishment Cycle (ESC)

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1 Introduction and Purpose

For a continuous review (Q, ROP) inventory system, customer demand during the replenishment lead time is stochastic. When lead-time demand exceeds the reorder point (ROP), a stockout occurs.

This document provides a mathematical derivation of the *Expected Shortage per Replenishment Cycle* (ESC). It details the integral calculus, infinite series expansion, uniform convergence justification via the Weierstrass M-test, and the explicit final analytical representation.

2 Variables, Assumptions, and Operational Logic

2.1 Assumptions and Parameter Definitions

The lead-time demand distribution is approximated by a normal distribution under the Central Limit Theorem, assuming independent and identically distributed demand across periods with finite mean and variance. The accuracy of this approximation depends on the underlying demand distribution and the number of aggregated demand periods. We define the foundational parameters and core assumptions:

- *Assumption (i.i.d. Demand):* Periodic customer demands are independent and identically distributed (i.i.d.) random variables with mean D and standard deviation σ_D.
- *Assumption (Constant Lead Time):* The replenishment lead time L is treated as a deterministic positive integer ($L \in \mathbb{N}^+$), representing the number of demand periods. Combined with the i.i.d. demand assumption, this ensures that lead-time demand variance is determined by the aggregation of demand uncertainty across periods.
- L: Replenishment lead time duration (number of periods).
- D_L: Expected demand during lead time, defined as $D_L = D \times L$.
- σ_L: Standard deviation of lead-time demand, aggregated as $\sigma_L = \sqrt{L}\sigma_D$.
- ss: Safety stock carried to buffer against lead-time demand uncertainty.
- ROP: Reorder point, defined as $\text{ROP} = D_L + ss$.
- z: Safety factor (z-score), standardizing safety stock:

$$z = \frac{\text{ROP} - D_L}{\sigma_L} = \frac{ss}{\sigma_L} \implies ss = z\sigma_L \tag{1}$$

  Note: Operationally, the safety factor satisfies $z \geq 0$ (since $ss \geq 0$), though the mathematical derivation holds generally for any $z \in \mathbb{R}$.
- $\phi(u)$: Standard normal probability density function (PDF), $\phi(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}u^2\right)$.
- $\Phi(u)$: Standard normal cumulative distribution function (CDF), $\Phi(u) = \int_{-\infty}^{u} \phi(t)\, dt$.

2.2 Expected Shortage Definition

A stockout in a given cycle occurs strictly when actual lead-time demand x exceeds ROP, with a shortage magnitude of $(x - \text{ROP})$. The Expected Shortage per Replenishment Cycle (ESC) is defined as the expected shortfall, expressed as the positive part expectation:

$$\text{ESC} = \mathbb{E}[(X - \text{ROP})^+] = \int_{\text{ROP}}^{\infty} (x - \text{ROP}) f(x)\, dx \tag{2}$$

Substituting $\text{ROP} = D_L + z\sigma_L$ and normal density $f(x)$:

$$\text{ESC} = \int_{D_L + z\sigma_L}^{\infty} [x - (D_L + z\sigma_L)] \cdot \frac{1}{\sigma_L \sqrt{2\pi}} \exp\left(-\frac{(x - D_L)^2}{2\sigma_L^2}\right) dx \tag{3}$$

3 Variable Transformation (u-Substitution)

We define the standardized normal variable u:

$$u = \frac{x - D_L}{\sigma_L} \implies x = D_L + u\sigma_L, \quad dx = \sigma_L\, du \tag{4}$$

Integration Limits: Lower bound z to upper bound ∞. Substituting into the ESC integral and factoring out σ_L:

$$\text{ESC} = \sigma_L \int_z^{\infty} (u - z)\phi(u)\, du \tag{5}$$

4 Integral Decomposition and Evaluation

Expanding the integrand yields two components:

$$\text{ESC} = \sigma_L \left[\int_z^{\infty} u\phi(u)\, du - z \int_z^{\infty} \phi(u)\, du \right] \tag{6}$$

4.1 First Term: $\int_z^{\infty} u\phi(u)\, du$ via PDF Derivative Property

Noting that $\phi'(u) = -u\phi(u)$, the first integral evaluates directly:

$$\int_z^{\infty} u\phi(u)\, du = -\int_z^{\infty} \phi'(u)\, du = -[\phi(u)]_z^{\infty} = 0 - (-\phi(z)) = \phi(z) \tag{7}$$

4.2 Second Term: Tail Probability via Maclaurin Series Expansion

The tail probability uses standard normal symmetry ($\int_0^{\infty} \phi(u)\, du = 0.5$):

$$\int_z^{\infty} \phi(u)\, du = 1 - \Phi(z) = 0.5 - \int_0^z \phi(u)\, du \tag{8}$$

Expressing $\exp(-\frac{1}{2}u^2)$ via its Maclaurin series ($\sum_{k=0}^{\infty} \frac{(-1)^k u^{2k}}{2^k k!}$) and integrating term-by-term over $[0, z]$ yields:

$$\int_0^z \phi(u)\, du = \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{2^k k!(2k+1)} = \frac{1}{\sqrt{2\pi}} \left(z - \frac{z^3}{6} + \frac{z^5}{40} - \frac{z^7}{336} + \dots \right) \tag{9}$$

5 Convergence Proof and Justification

To validate absolute convergence, we apply d'Alembert's Ratio Test to absolute terms $a_k = \frac{u^{2k}}{2^k k!}$:

$$\rho = \lim_{k\to\infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k\to\infty} \frac{u^2}{2(k+1)} = 0 \tag{10}$$

Since $\rho = 0 < 1$ for all finite $u \in \mathbb{R}$, the exponential series exhibits **absolute convergence** everywhere ($R = \infty$).

Furthermore, to justify term-by-term integration, we invoke the *Weierstrass M-test*. On any compact interval $[0, M]$, every term is bounded by:

$$\left| \frac{(-1)^k u^{2k}}{2^k k!} \right| \leq \frac{M^{2k}}{2^k k!} := M_k \tag{11}$$

Since the series $\sum_{k=0}^{\infty} M_k = e^{M^2/2} < \infty$, the series converges uniformly on $[0, M]$ by the Weierstrass M-test, validating the legitimacy of term-by-term integration.

Since every finite real interval is compact, the same argument applies to any compact interval in $\mathbb{R}$. Therefore, term-by-term integration is justified for every finite real value of z.

6 Standard Normal Loss Function and Final Formulas

We formally define the Standard Normal Loss Function $G(z)$ as:

$$G(z) := \int_z^{\infty} (u - z)\phi(u)\, du = \phi(z) - z(1 - \Phi(z)) \tag{12}$$

Multiplying by σ_L yields the standard compact formulation:

$$\text{ESC} = \sigma_L \cdot G(z) = \sigma_L \left[\phi(z) - z(1 - \Phi(z))\right] \tag{13}$$

Alternatively, substituting the explicit Maclaurin series expansion of the tail probability, we obtain the following infinite-series representation:

$$\text{ESC} = \sigma_L \phi(z) - \sigma_L z \left[0.5 - \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{2^k k!(2k+1)} \right] \tag{14}$$

Expressed directly in terms of safety stock ($z = \frac{ss}{\sigma_L}$):

$$\text{ESC} = \sigma_L \phi\left(\frac{ss}{\sigma_L}\right) - ss \left[0.5 - \frac{1}{\sqrt{2\pi}} \left(\frac{ss}{\sigma_L} - \frac{1}{6}\left(\frac{ss}{\sigma_L}\right)^3 + \frac{1}{40}\left(\frac{ss}{\sigma_L}\right)^5 - \dots \right) \right] \tag{15}$$